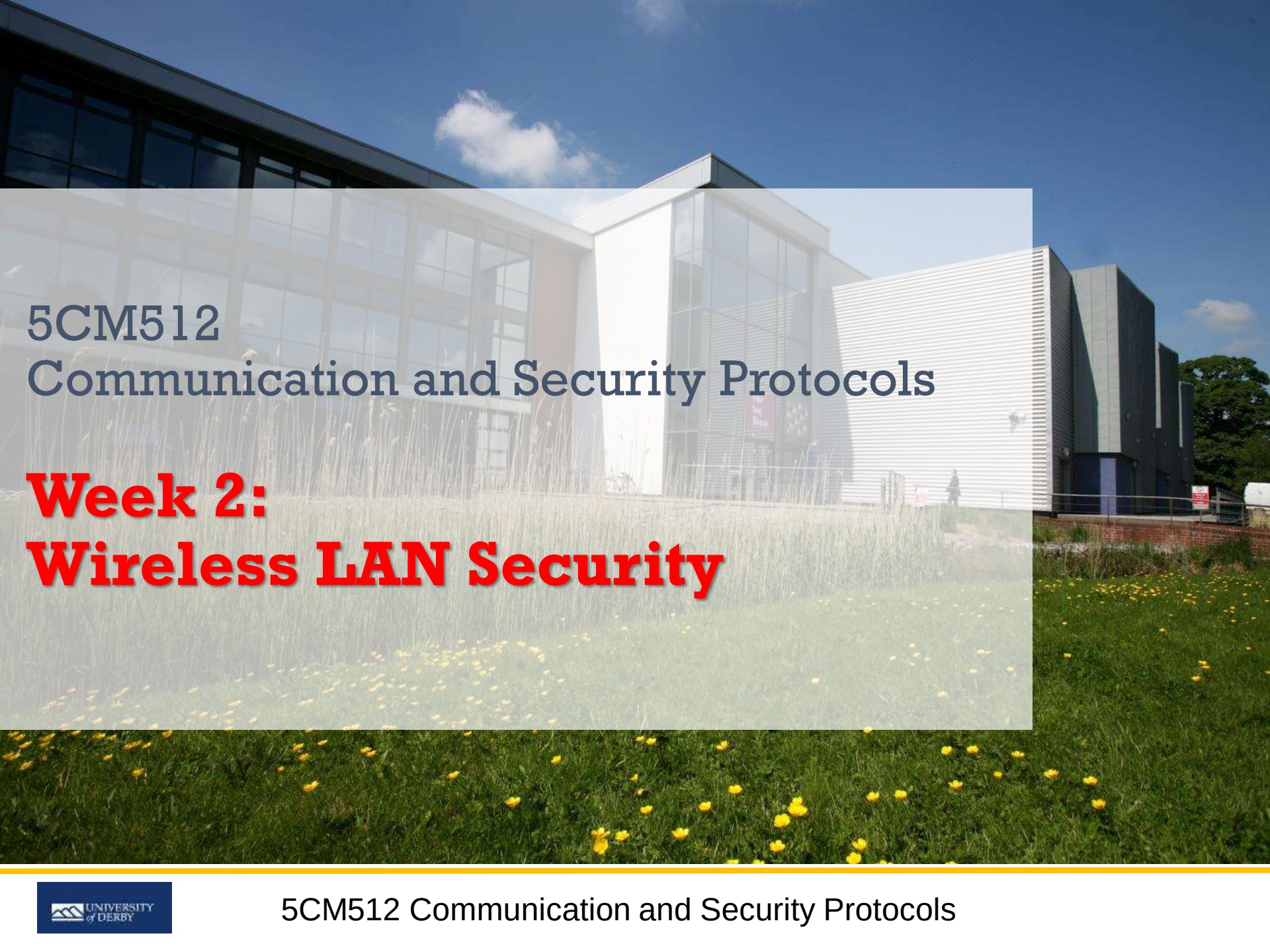




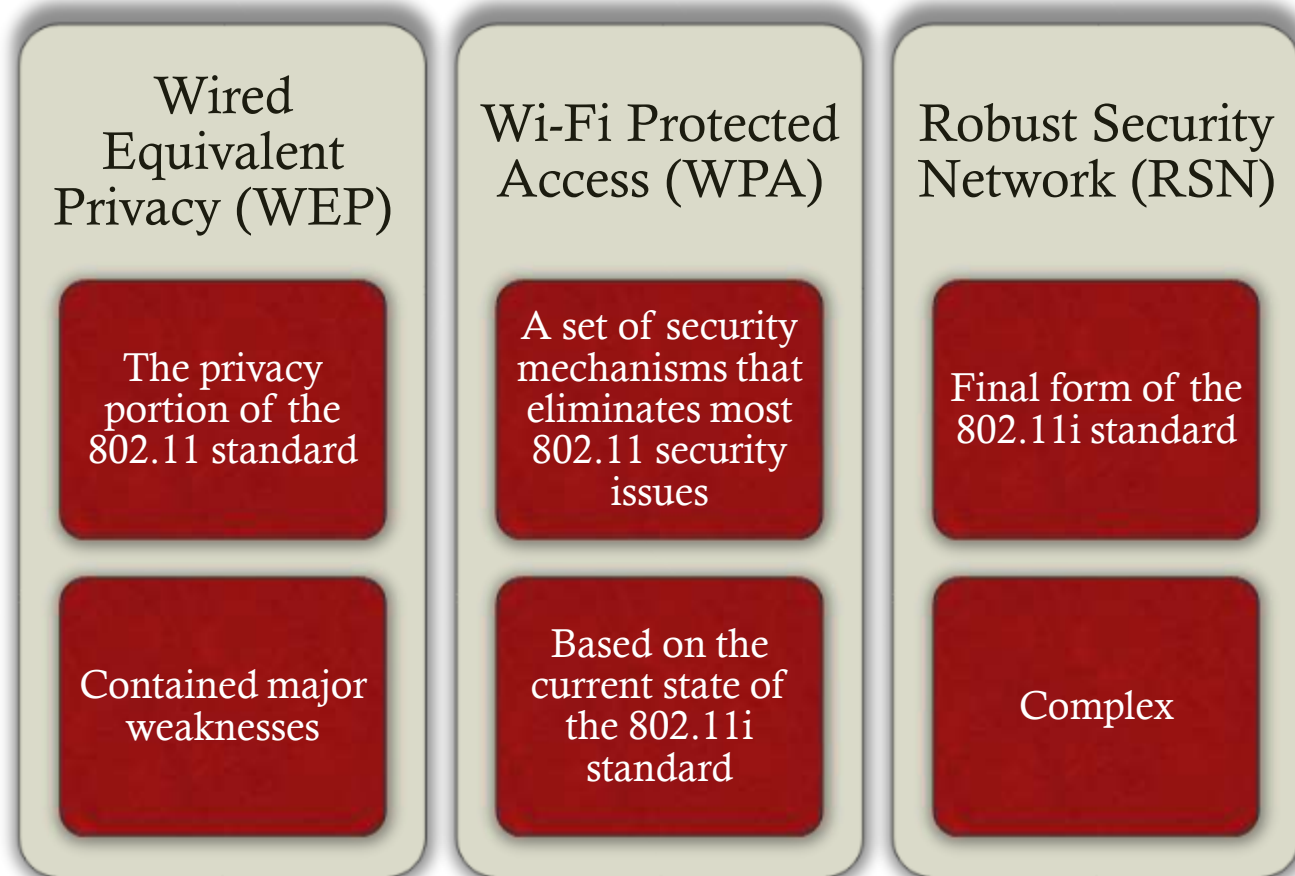
5CM512 Communication and Security Protocols

Week 2: Wireless LAN Security

IEEE 802.11i Wireless LAN Security

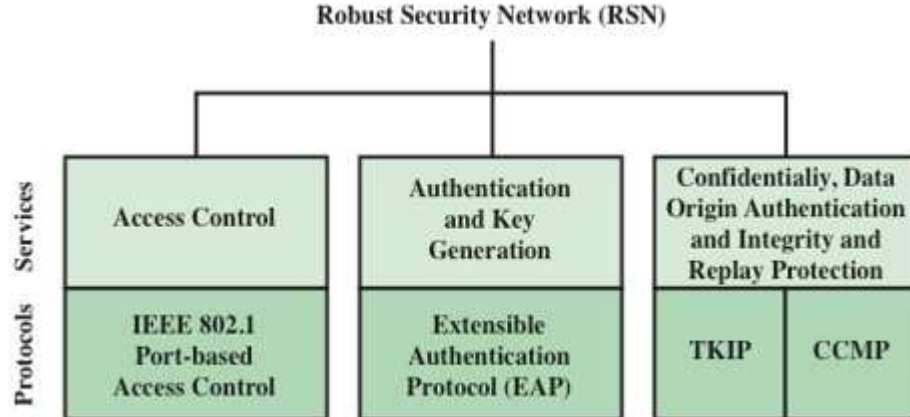
IEEE 802.11, The Working Group Setting the Standards for Wireless LANs

- There is an increased need for robust security services and mechanisms for wireless LANs

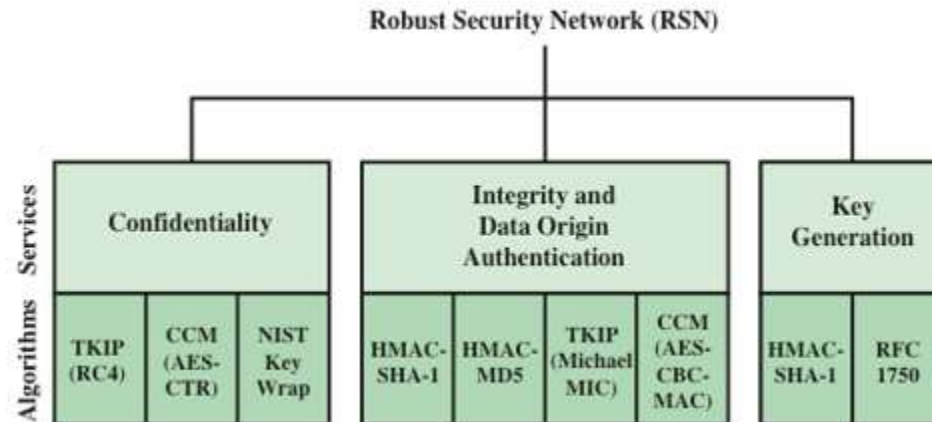


Elements of IEEE 802.11i

- Services and Protocols

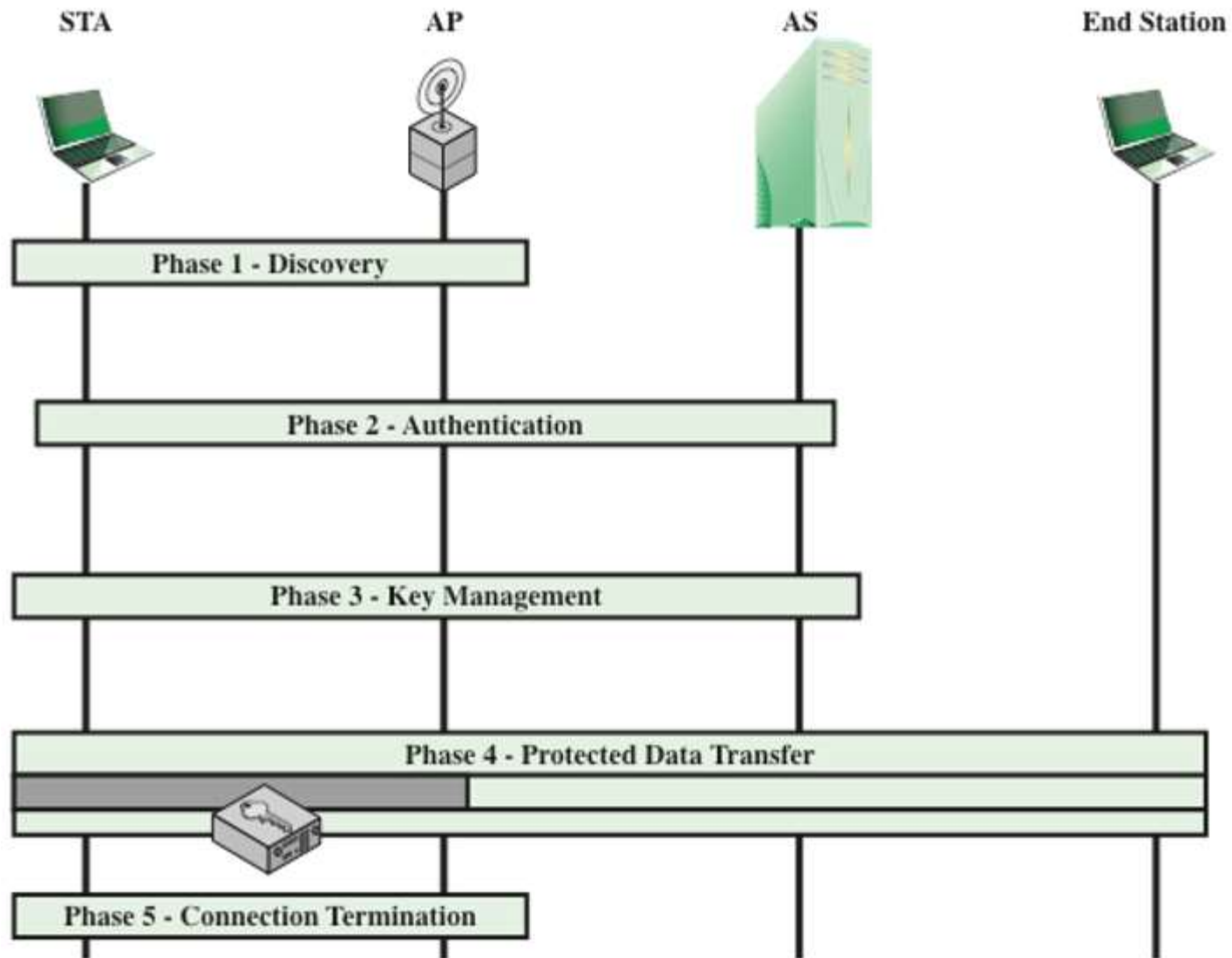


- Cryptographic Algorithms

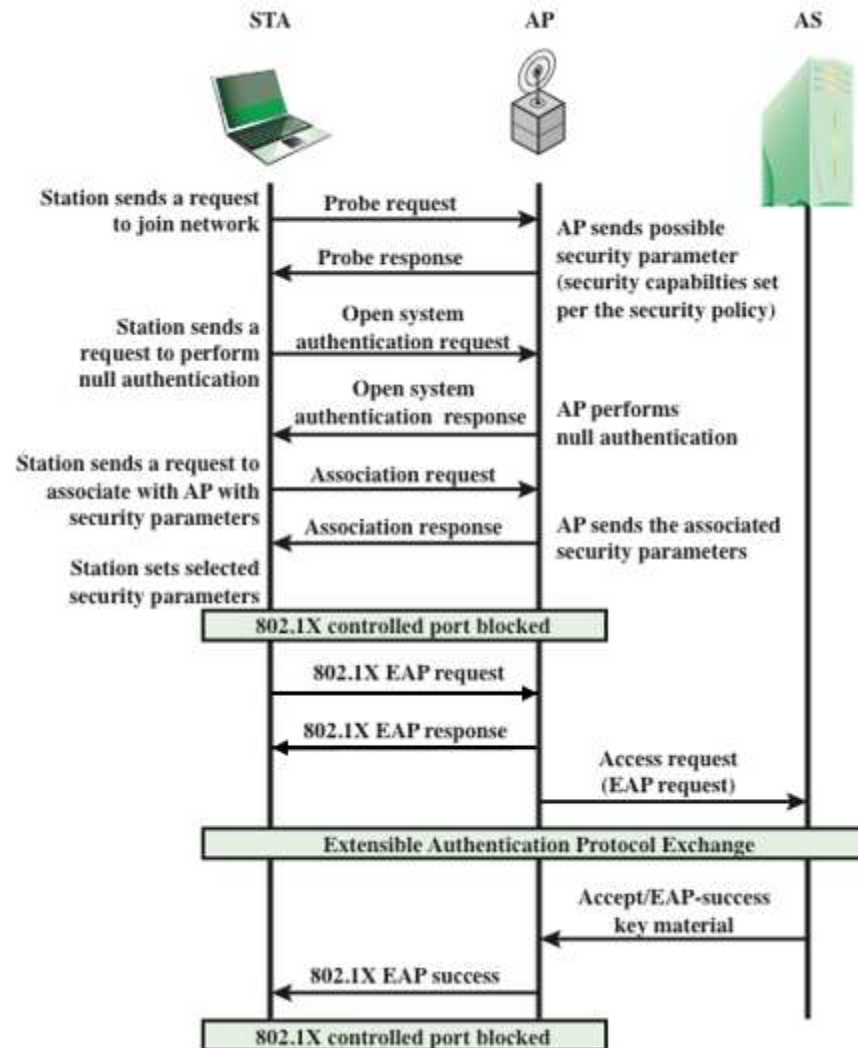


CBC-MAC = Cipher Block Chaining Message Authentication Code (MAC)
 CCM = Counter Mode with Cipher Block Chaining Message Authentication Code
 CCMP = Counter Mode with Cipher Block Chaining MAC Protocol
 TKIP = Temporal Key Integrity Protocol

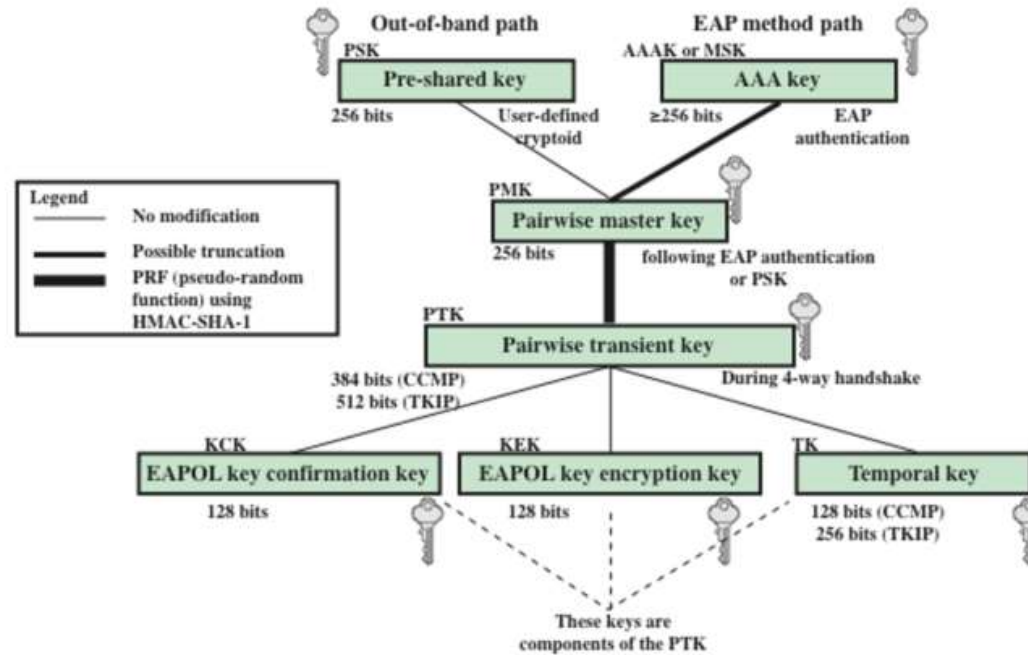
IEEE 802.11i Phase of Operation



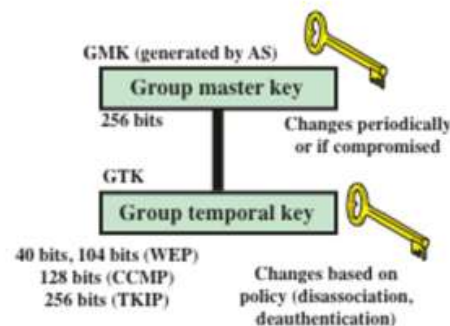
IEEE 802.11i Phases of Operation : Capability Discovery, Authentication and Association



IEEE 802.11i Key Hierarchies



(a) Pairwise key hierarchy

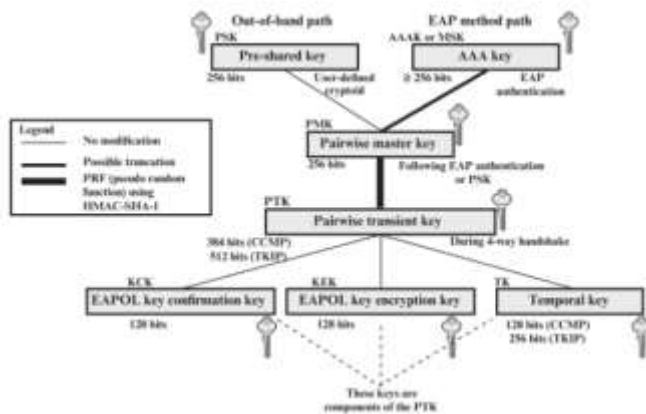


(b) Group key hierarchy

IEEE 802.11i Keys for Data Confidentiality and Integrity Protocols

Abbreviation	Name	Description / Purpose	Size (bits)	Type
AAA Key	Authentication, Accounting, and Authorization Key	Used to derive the PMK. Used with the IEEE 802.1X authentication and key management approach. Same as MMSK.	≥ 256	Key generation key, root key
PSK	Pre-shared Key	Becomes the PMK in pre-shared key environments.	256	Key generation key, root key
PMK	Pairwise Master Key	Used with other inputs to derive the PTK.	256	Key generation key
GMK	Group Master Key	Used with other inputs to derive the GTK.	128	Key generation key
PTK	Pair-wise Transient Key	Derived from the PMK. Comprises the EAPOL-KCK, EAPOL-KEK, and TK and (for TKIP) the MIC key.	512 (TKIP) 384 (CCMP)	Composite key
TK	Temporal Key	Used with TKIP or CCMP to provide confidentiality and integrity protection for unicast user traffic.	256 (TKIP) 128 (CCMP)	Traffic key
GTK	Group Temporal Key	Derived from the GMK. Used to provide confidentiality and integrity protection for multicast/broadcast user traffic.	256 (TKIP) 128 (CCMP) 40,104 (WEP)	Traffic key
MIC Key	Message Integrity Code Key	Used by TKIP's Michael MIC to provide integrity protection of messages.	64	Message integrity key
EAPOL-KCK	EAPOL-Key Confirmation Key	Used to provide integrity protection for key material distributed during the 4-Way Handshake.	128	Message integrity key
EAPOL-KEK	EAPOL-Key Encryption Key	Used to ensure the confidentiality of the GTK and other key material in the 4-Way Handshake.	128	Traffic key / key encryption key
WEP Key	Wired Equivalent Privacy Key	Used with WEP.	40,104	Traffic key

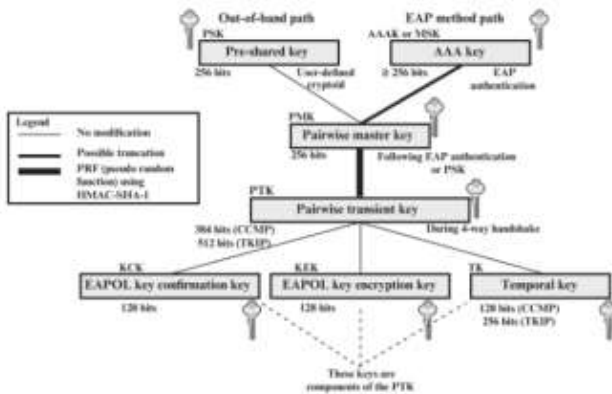
Pairwise Keys



- **Used for communication between a pair of devices, typically between a STA and an AP**
 - These keys form a hierarchy beginning with a master key from which other keys are derived dynamically and used for a limited period of time
- **Pre-shared key (PSK)**
 - A secret key shared by the AP and a STA and installed in some fashion outside the scope of IEEE 802.11i
- **Master session key (MSK)**
 - Also known as the AAAK, and is generated using the IEEE 802.1X protocol during the authentication phase
- **Pairwise master key (PMK)**
 - Derived from the master key
 - If a PSK is used, then the PSK is used as the PMK; if a MSK is used, then the PMK is derived from the MSK by truncation
- **Pairwise transient key (PTK)**
 - Consists of three keys to be used for communication between a STA and AP after they have been mutually authenticated
 - Using the STA and AP addresses in the generation of the PTK provides protection against session hijacking and impersonation; using nonces provides additional random keying material

PTK Parts

- The three parts of the PTK are:



EAP Over LAN (EAPOL) Key Confirmation Key (EAPOL-KCK)

- Supports the integrity and data origin authenticity of STA-to-AP control frames during operational setup of an RSN
- It also performs an access control function: proof-of-possession of the PMK
- An entity that possesses the PMK is authorized to use the link

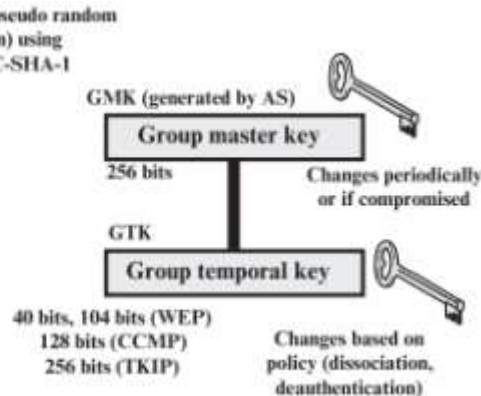
EAPOL Key Encryption Key (EAPOL-KEK)

- Protects the confidentiality of keys and other data during some RSN association procedures

Temporal Key (TK)

- Provides the actual protection for user traffic

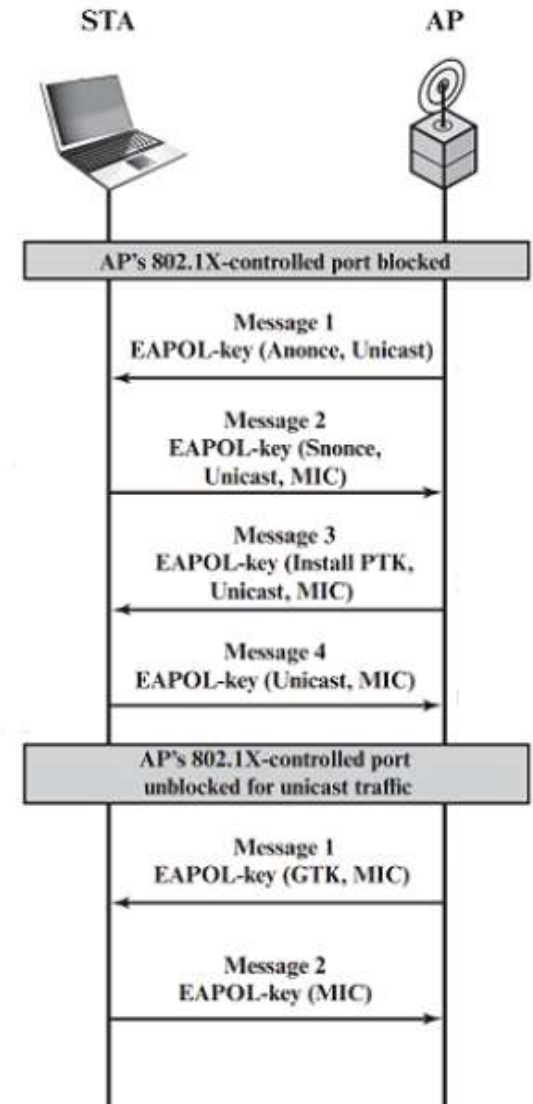
Group Keys

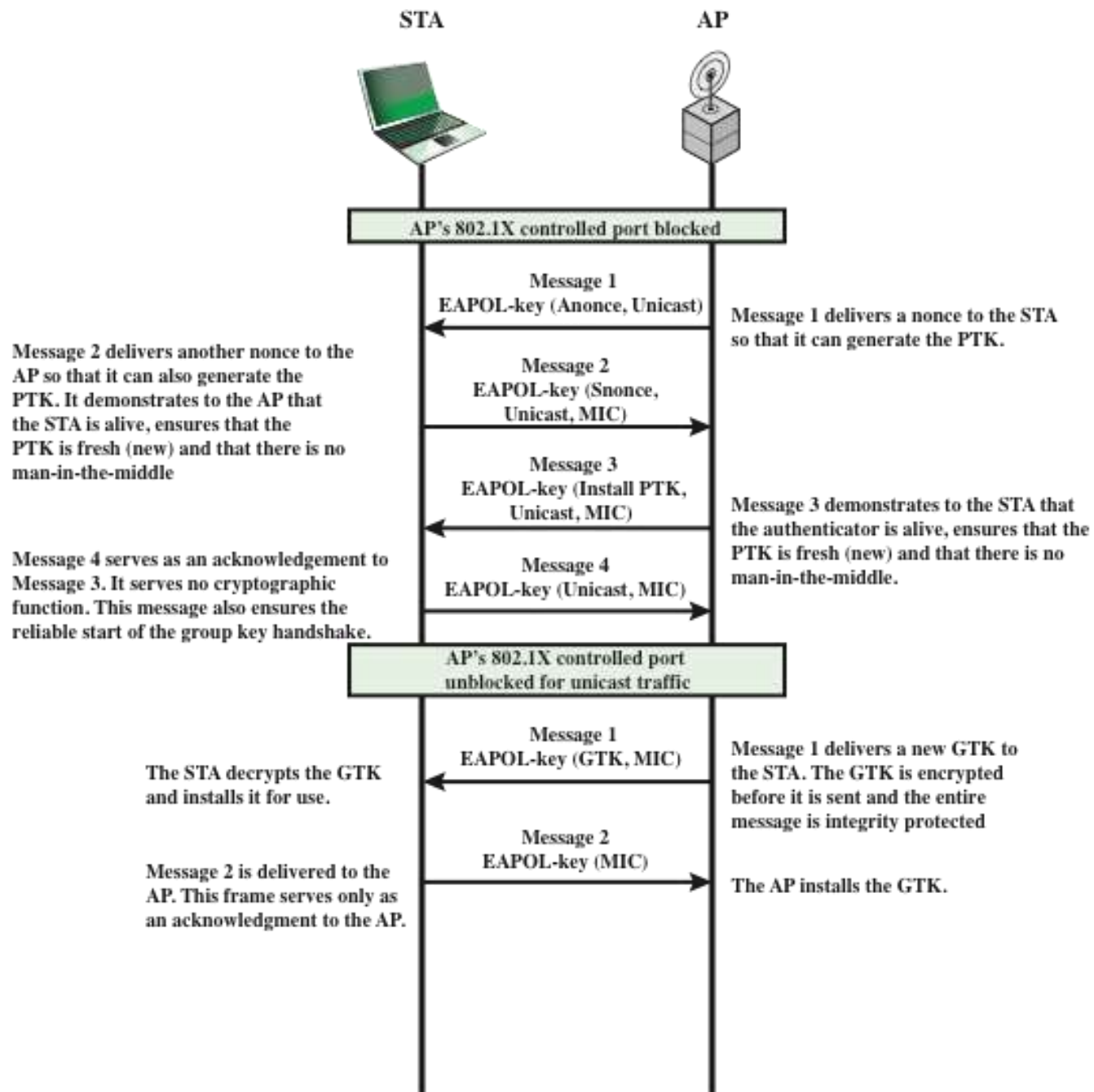


- Group keys are used for multicast communication in which one STA sends MPDUs to multiple STAs
 - Group master key (GMK)
 - Key-generating key used with other inputs to derive the GTK
 - Group temporal key (GTK)
 - Generated by the AP and transmitted to its associated STAs
 - IEEE 802.11i requires that its value is computationally indistinguishable from random
 - Distributed securely using the pairwise keys that are already established
 - Is changed every time a device leaves the network

Class Discussion

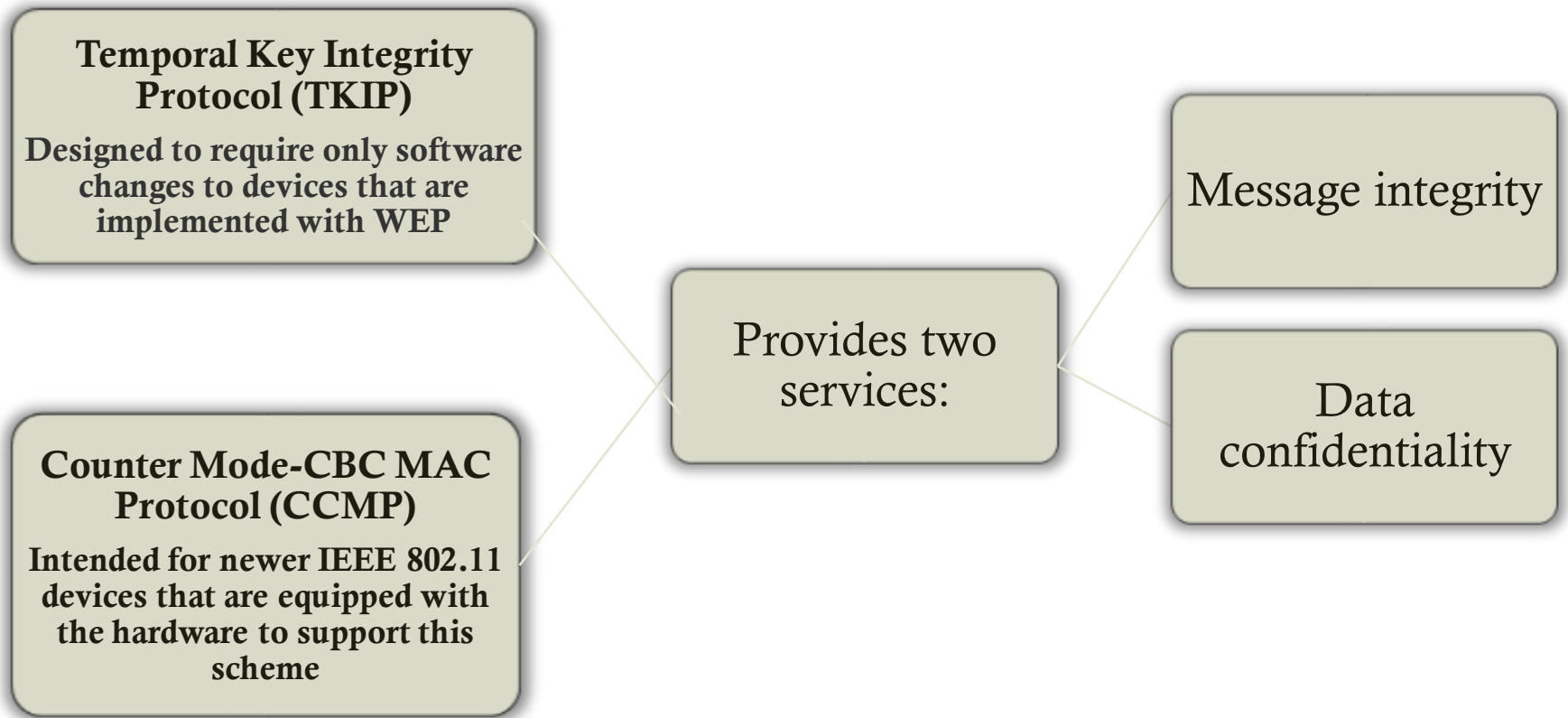
- How can we implement pairwise key distribution?
 - This is called 4-Way Handshake
- Discuss the messages in 4-Way Handshake and Group Key Handshake
 - What messages are designed for against Man in the Middle attack?
In what way?





Protected Data Transfer Phase

- IEEE 802.11i defines two schemes for protecting data transmitted in 802.11 MPDUs:

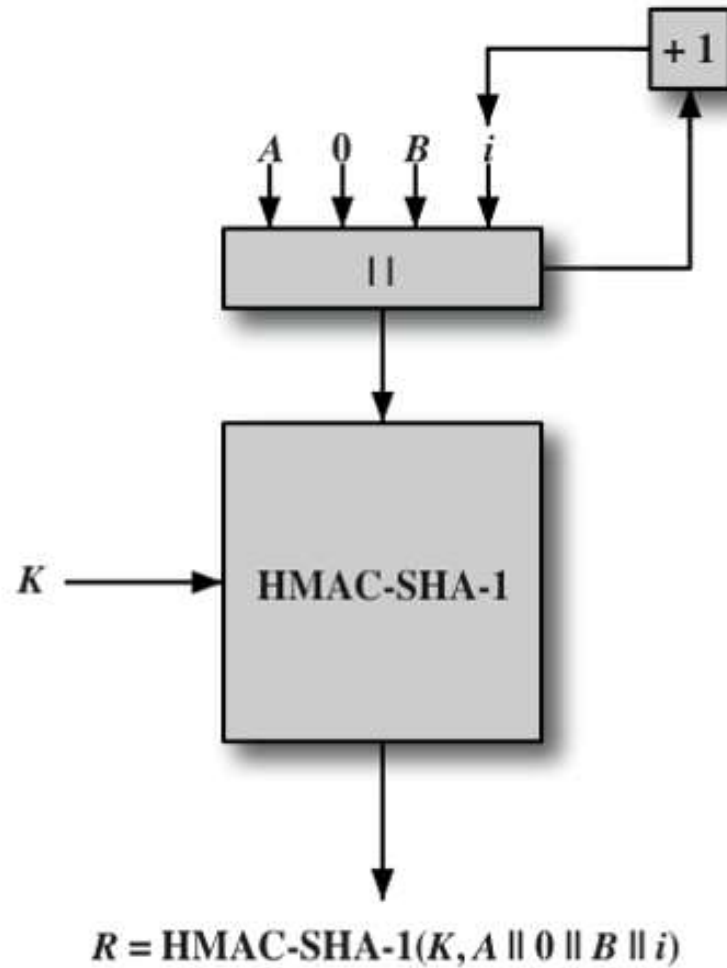


IEEE 802.11i

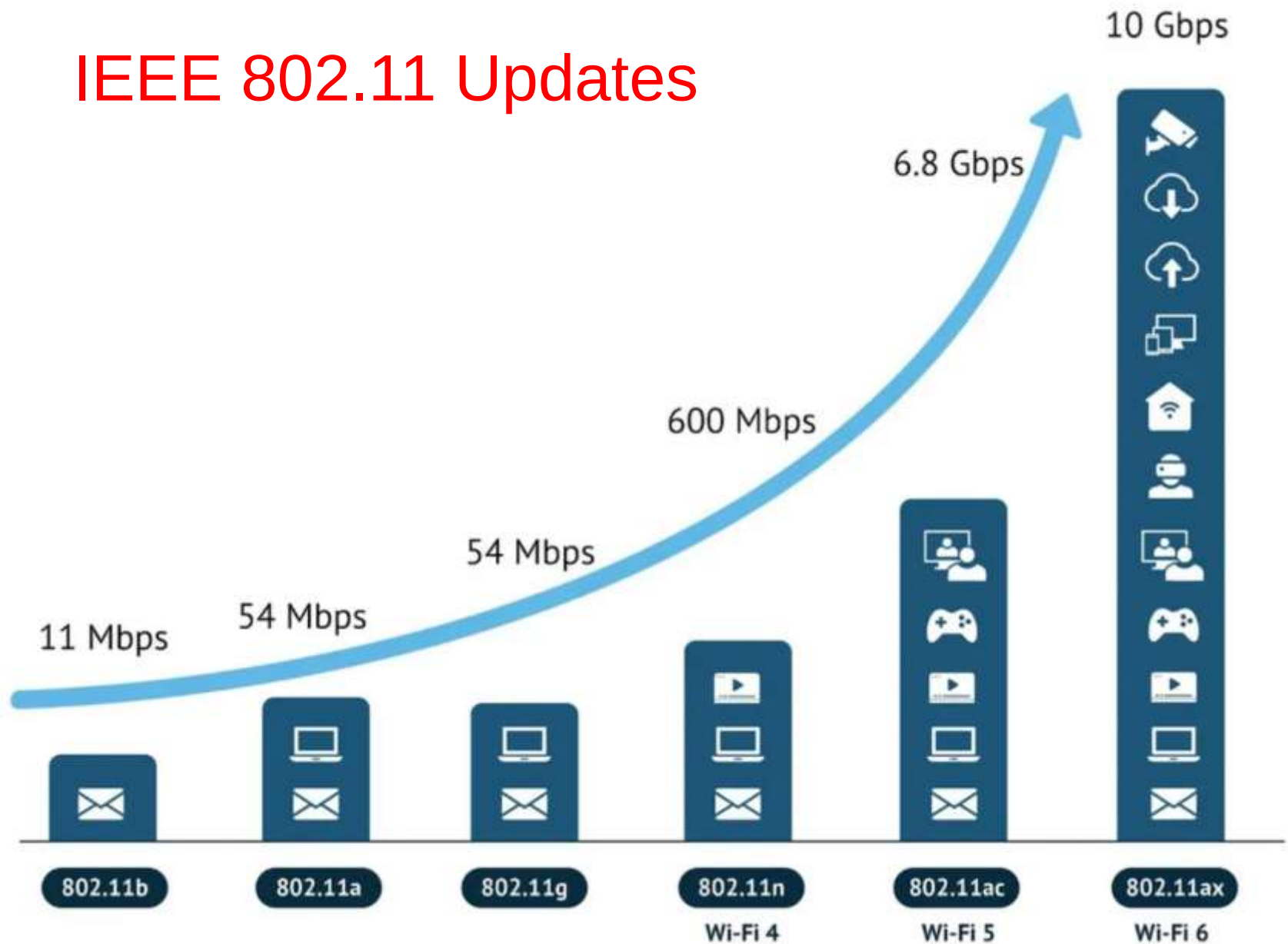
Pseudorandom Function (PRF)

- Used at a number of places in the IEEE 802.11i scheme (to generate nonces, to expand pairwise keys, to generate the GTK)
 - Best security practice dictates that different pseudorandom number streams be used for these different purposes
- Built on the use of HMAC-SHA-1 to generate a pseudorandom bit stream

IEEE 802.11i Pseudorandom Function



IEEE 802.11 Updates



IEEE 802.11 Protocol	Release Date	Frequency Band(s)	Bandwidth	Max Throughput
802.11-1997	1997	2.4	22	2 Mbps
11b	1999	2.4	22	11 Mbps
11a	1999	5	20	54 Mbps
11g	2003	2.4	20	54 Mbps
11n (Wi-Fi 4)	2009	2.4/5	20/40	600 Mbps
11ac (Wi-Fi 5)	2013	5	20/40/80/160	6.8 Gbps
11ax (Wi-Fi 6)	2019	2.5/5	20/40/80/160	9.6 Gbps
11ax (WiFi 6E)	2020	2.5/5/6	20/40/80/160	9.6 Gbps
11be (WiFi 7)	2024 (expected)	2.5/5/6	20/40/80/160/320	46.1 Gbps

Any QUESTIONS?

